Auteur: Stan Van Campenhout
Studierichting: Informatica
Oefeningenreeks 4A nr. 18.6

$$\int \left(x^3 - x^2\right) \cos x dx = \int \left(x^3 - x^2\right) d\left(\sin x\right) = \left(x^3 - x^2\right) \sin x - \int \sin x d\left(x^3 - x^2\right)$$
$$= \left(x^3 - x^2\right) \sin x - \int \left(3x^2 - 2x\right) \sin x dx$$
$$= \left(x^3 - x^2\right) \sin x + \left(3x^2 - 2x\right) \cos x - \int \left(6x - 2\right) \cos x dx$$
$$= \left(x^3 - x^2\right) \sin x + \left(3x^2 - 2x\right) \cos x - \int 6x \cos x dx + 2 \int \cos x dx$$
$$= \left(x^3 - x^2\right) \sin x + \left(3x^2 - 2x\right) \cos x - 6x \sin x + \int \sin x d\left(6x\right) + 2 \sin x$$
$$= \left(x^3 - x^2\right) \sin x + \left(3x^2 - 2x\right) \cos x - 6x \sin x - 6 \cos x + 2 \sin x + c$$

References